

11.15

(a) By definition, $SSE = \sum (y_i - \hat{y})^2$

$$= \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 \quad (\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x)$$

$$= \sum ((y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x}))^2 \quad (\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x})$$

$$= \sum (y_i - \bar{y})^2 + \hat{\beta}_1^2 \sum (x_i - \bar{x})^2$$

$$- 2 \sum \hat{\beta}_1 (y_i - \bar{y})(x_i - \bar{x})^2$$

$$\left(\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} \right) = \sum (y_i - \bar{y})^2 - \hat{\beta}_1 \sum (x_i - \bar{x})(y_i - \bar{y})$$

$$(\text{By definition}) = S_{YY} - \hat{\beta}_1 S_{XY}.$$

$$(b) \quad \hat{\beta}_1 S_{XY} = \frac{S_{XY}^2}{S_{XX}} \leq 0, \text{ so } SSE \leq S_{YY}.$$

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The Likelihood function is

$$L(\beta_0, \beta_1, \sigma^2) = \left(\frac{1}{\sqrt{2\pi}\sigma} \right)^n e^{-\frac{1}{2\sigma^2} \sum (y_i - \beta_0 - \beta_1 x_i)^2}$$

log likelihood

$$\ell(\beta_0, \beta_1, \sigma^2) = -n \ln \sigma - \frac{1}{2\sigma^2} \sum (y_i - \beta_0 - \beta_1 x_i)^2.$$

$$\text{Hence } \hat{\beta}_0, \hat{\beta}_1 = \arg \max_{\beta_0, \beta_1} \sum (y_i - \beta_0 - \beta_1 x_i)^2,$$

which is exactly the solution of OLS.

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$$\frac{\partial \ell}{\partial \sigma^2} = 0 \Rightarrow -\frac{n}{2\sigma^2} + \frac{SSE}{2\sigma^4} = 0$$

$$\therefore \hat{\sigma}^2 = \frac{SSE}{n}$$

11.30

(a) For small-particle case,

$$\text{slope } \hat{\beta}_1 = 0.155, n=31, V(\hat{\beta}_1) = (0.0202)^2,$$

$$SSE = 2.04, \alpha = 0.05.$$

$$H_0: \hat{\beta}_1 = 0, H_1: \hat{\beta}_1 \neq 0.$$

$$\text{Test statistic is } T = \frac{\hat{\beta}_1}{\sqrt{V(\hat{\beta}_1)}} = 7.67.$$

$$\text{Critical value is } t_{n-1; \frac{\alpha}{2}} = 2.04$$

$T > t_{n-1; \frac{\alpha}{2}}$, so we reject the null.

The test for large-particle is similar.